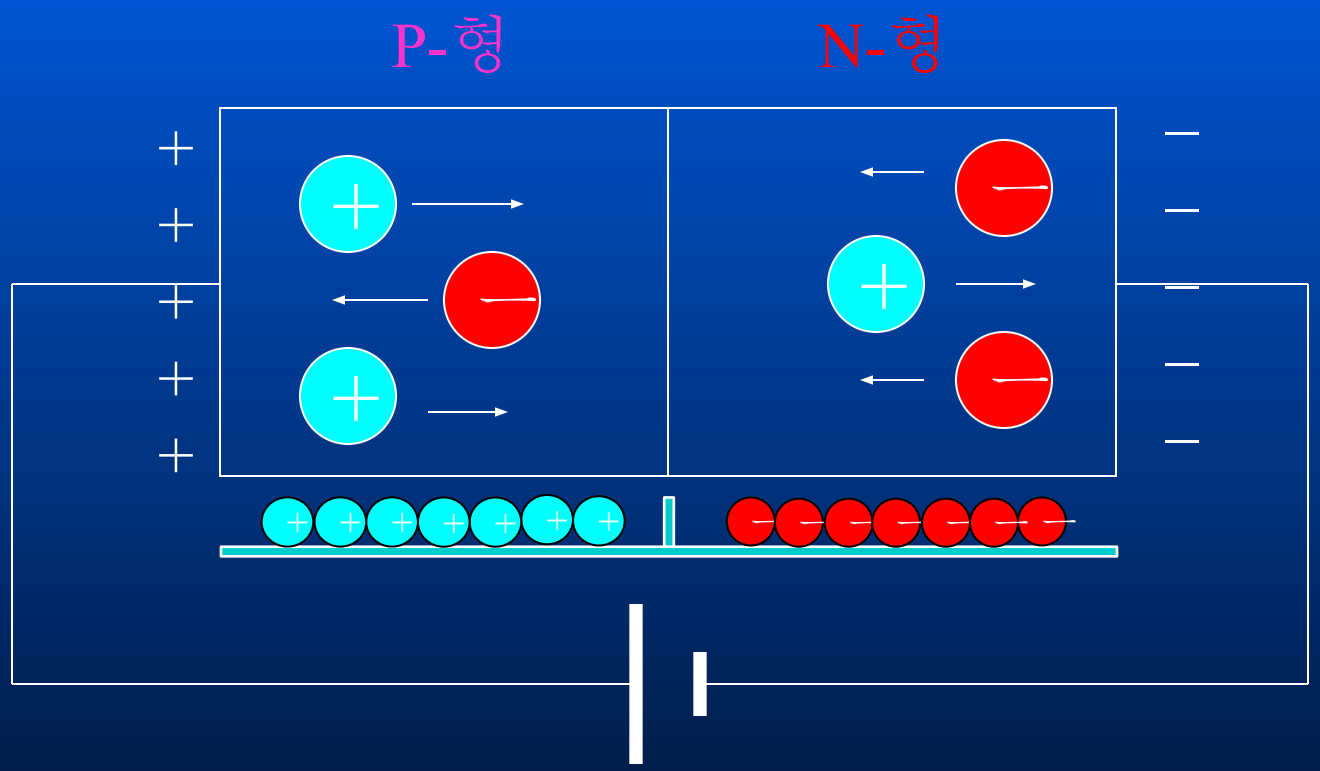
1장 전력전자 소자

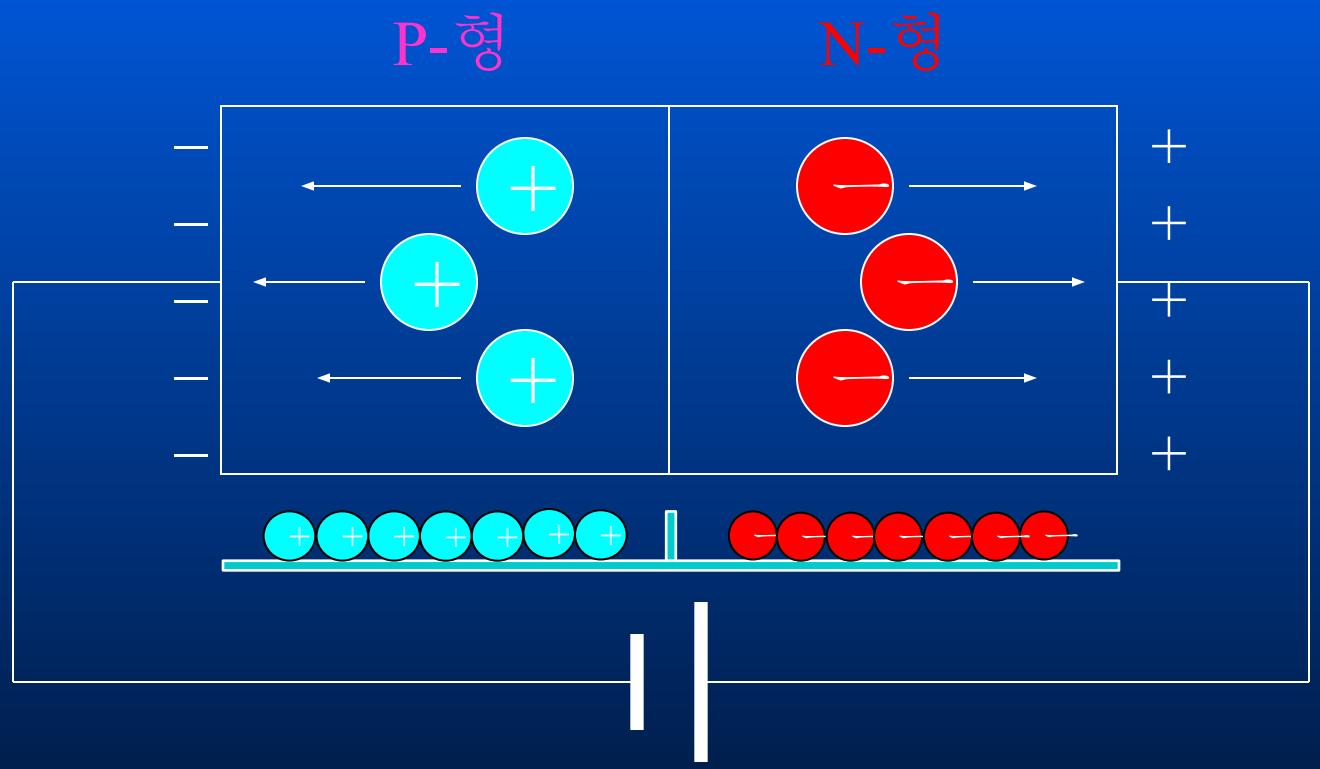
Power Electronics Components

1-1. Diode의 동작원리: 순방향 (Forward)



- 전류 = 전자전류 + 정공전류
- 전자는 전류와 반대 방향
- 정공은 전류와 같은 방향

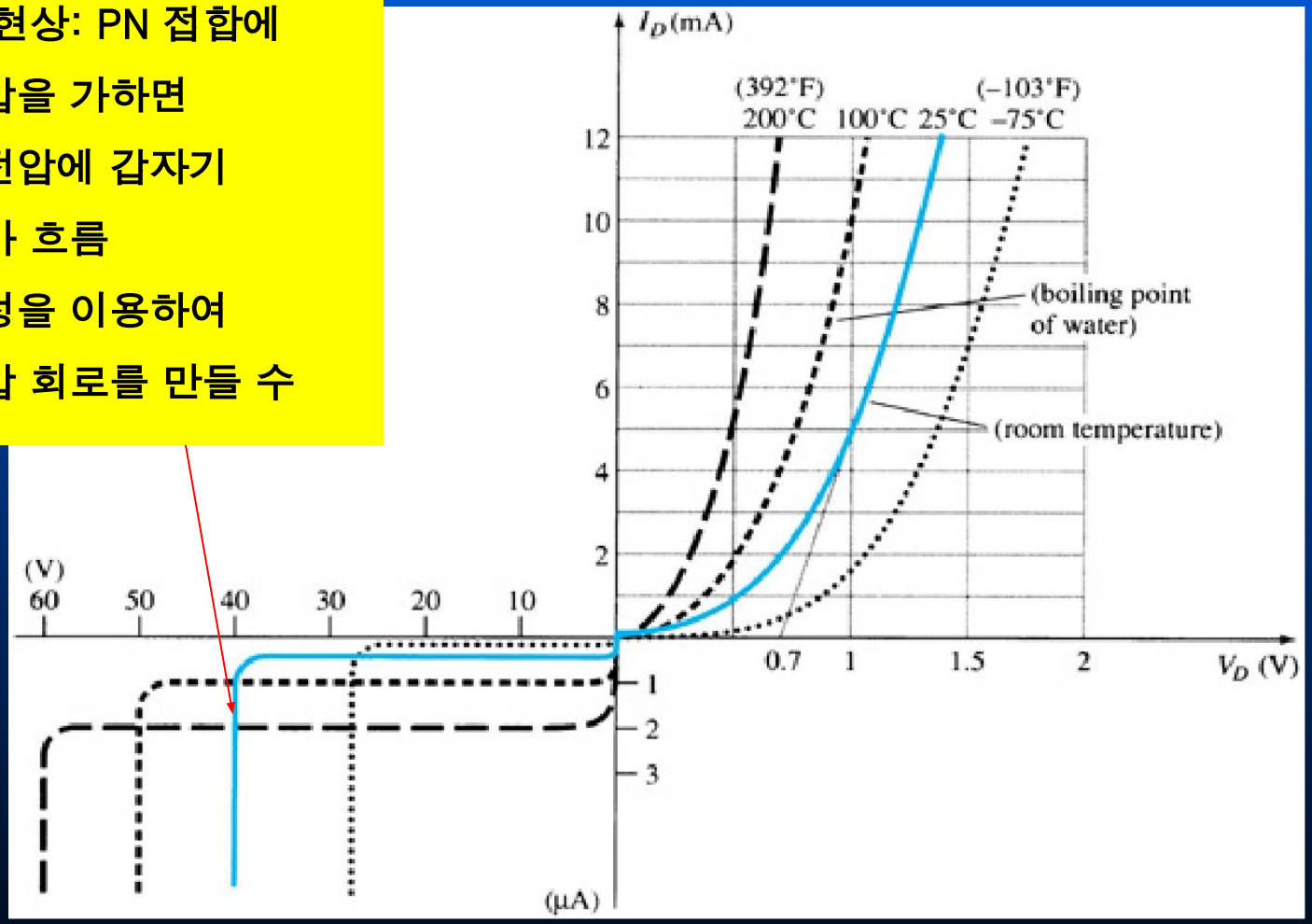
1-2. Diode의 동작원리: 역방향 (Reverse)



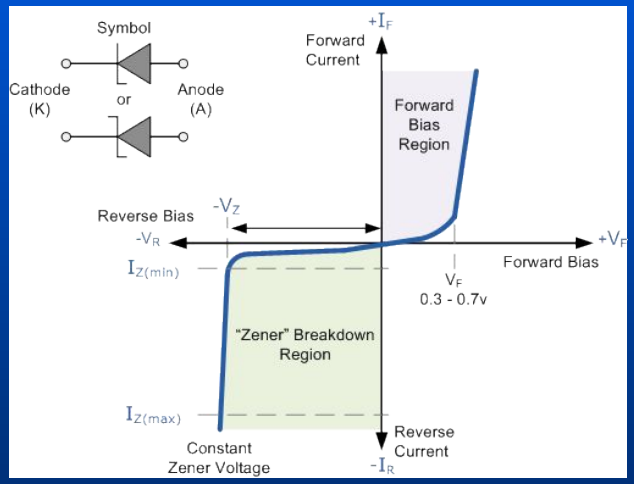
- 전류 = 전자전류 + 정공전류
- 전류가 흐르지 못함

1-4. Diode의 동작원리: 특성 곡선

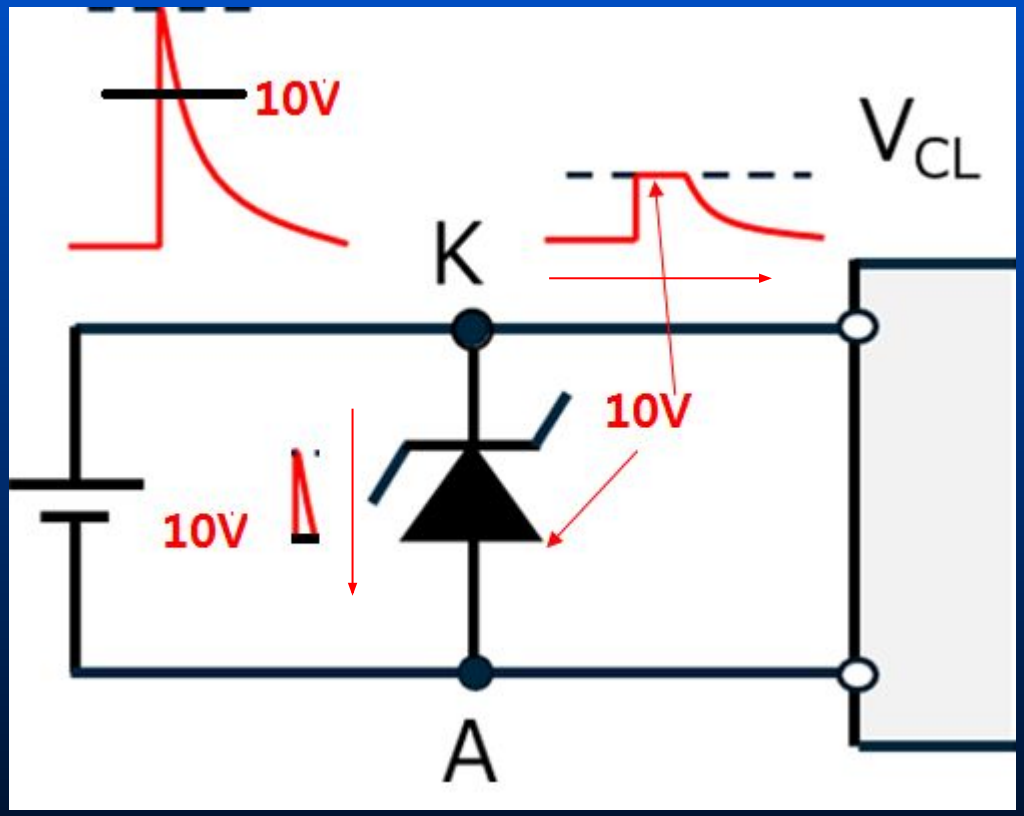
1. 제너 현상: PN 접합에 역전압을 가하면 특정전압에 갑자기 전류가 흐름
2. 이특성을 이용하여 정전압 회로를 만들 수 있음



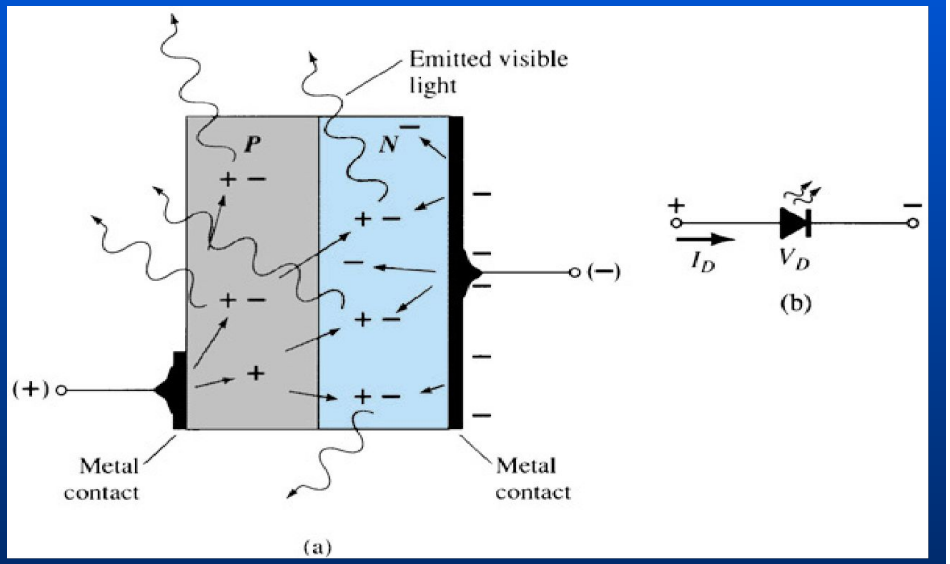
1-6. Diode의 동작원리: Zener Diode의 동작



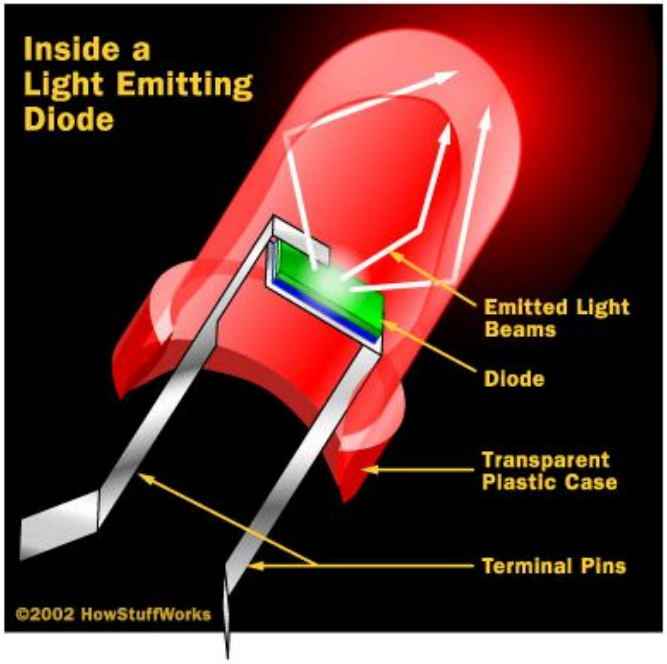
1. 제너 현상: PN 접합에 역전압을 가하면 특정전압에 갑자기 전류가 흐름
2. 이특성을 이용하여 정전압 회로를 만들 수 있음



1-5. Diode의 동작원리: LED의 동작

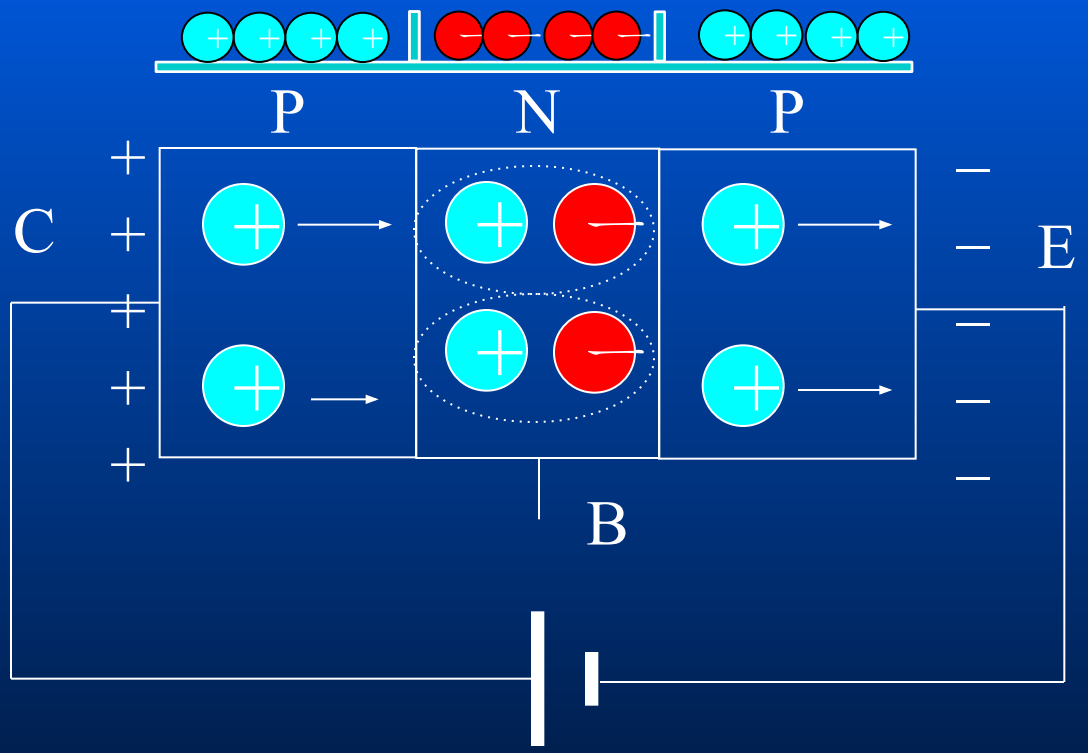


1. Light Emitting Diode
2. PN 접합을 통과시 빛이 발생하는 다이오드
3. Red, Green, Blue가 있고 RGB을 합하면 White인 백색 LED



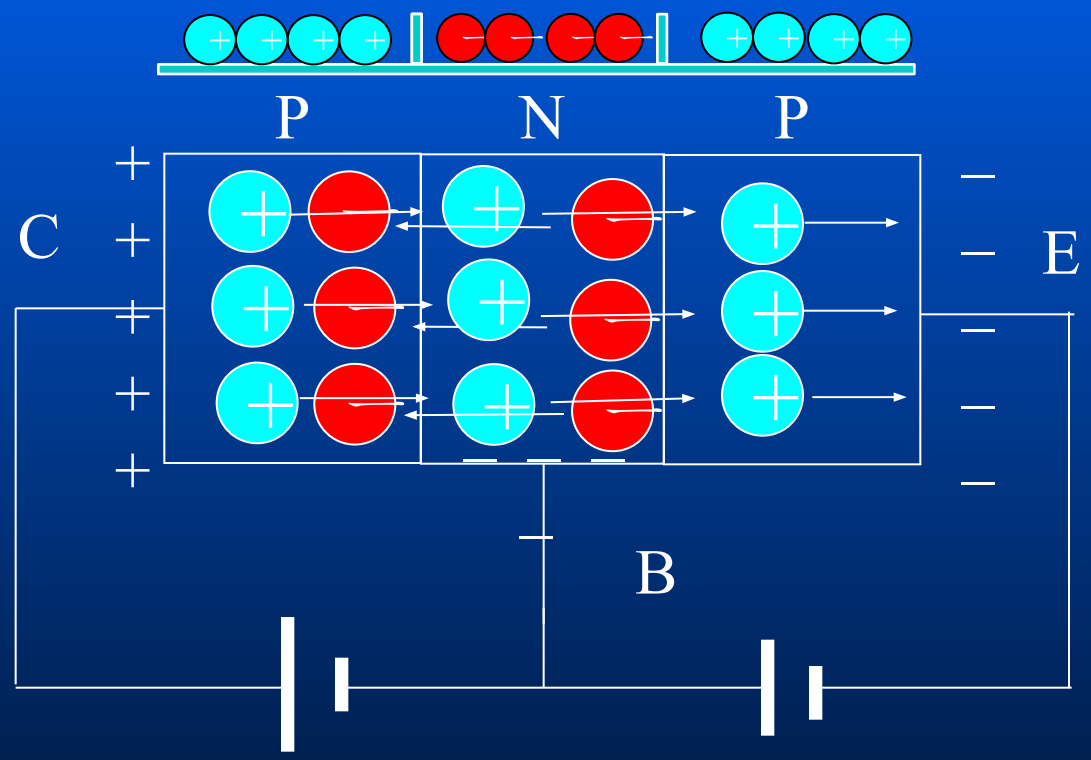
LEDs have several advantages over conventional incandescent lamps. For one thing, they don't have a filament that will burn out, so they last much longer. Additionally, their small plastic bulb makes them a lot more durable. They also fit more easily into modern electronic circuits.

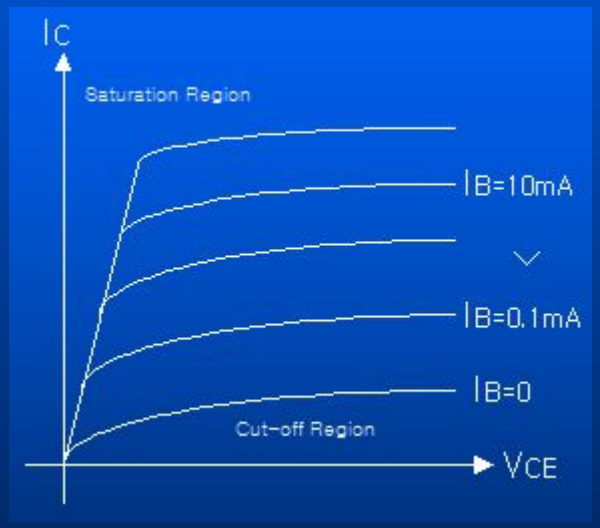
2-1. BJT(Bipolar Junction Transistor)



- 전류가 흐르지 못함
- 고전압의 경우는 Brake down 전류가 흐름

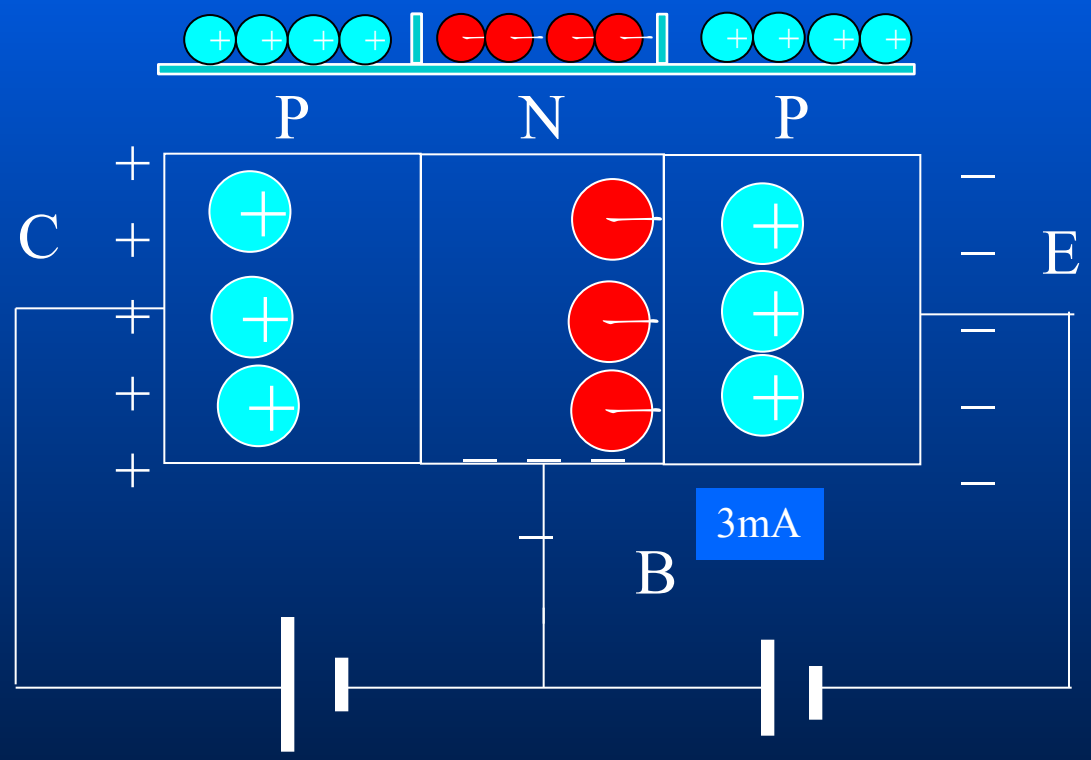
2-2. BJT의 동작원리: Bias 인가

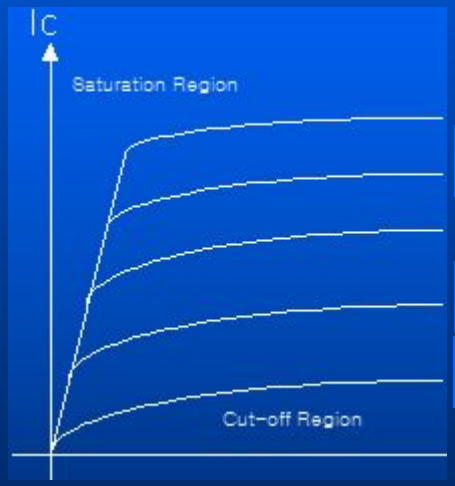




- Base에 전류가 인가되면 Emitter로 +가 이동
- N에 +가 이동
- 전체적으로 전류가 흐름

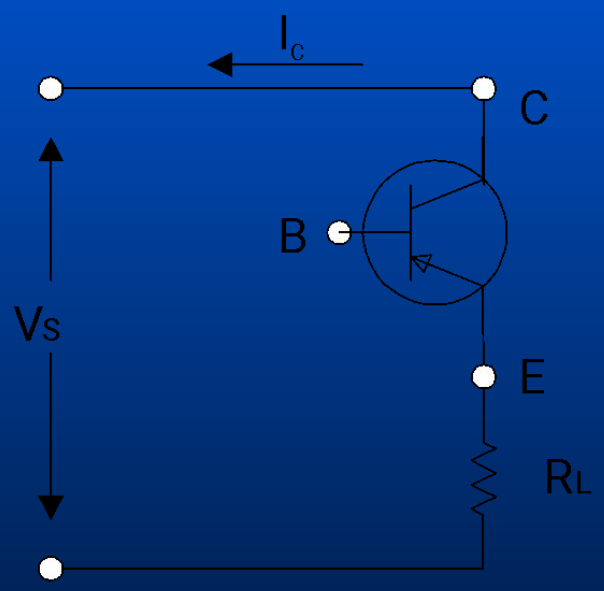
2-2. BJT의 동작원리: Bias 인가



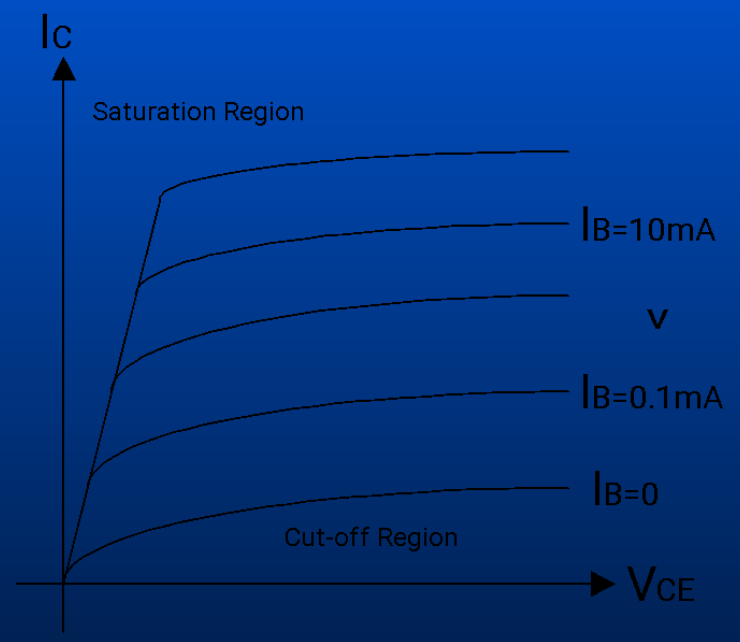


- Base에 전류가 인가되면 Emitter로 +가 이동
- N에 +가 이동
- 전체적으로 전류가 흐름

2-4. BJT의 동작원리: 특성곡선

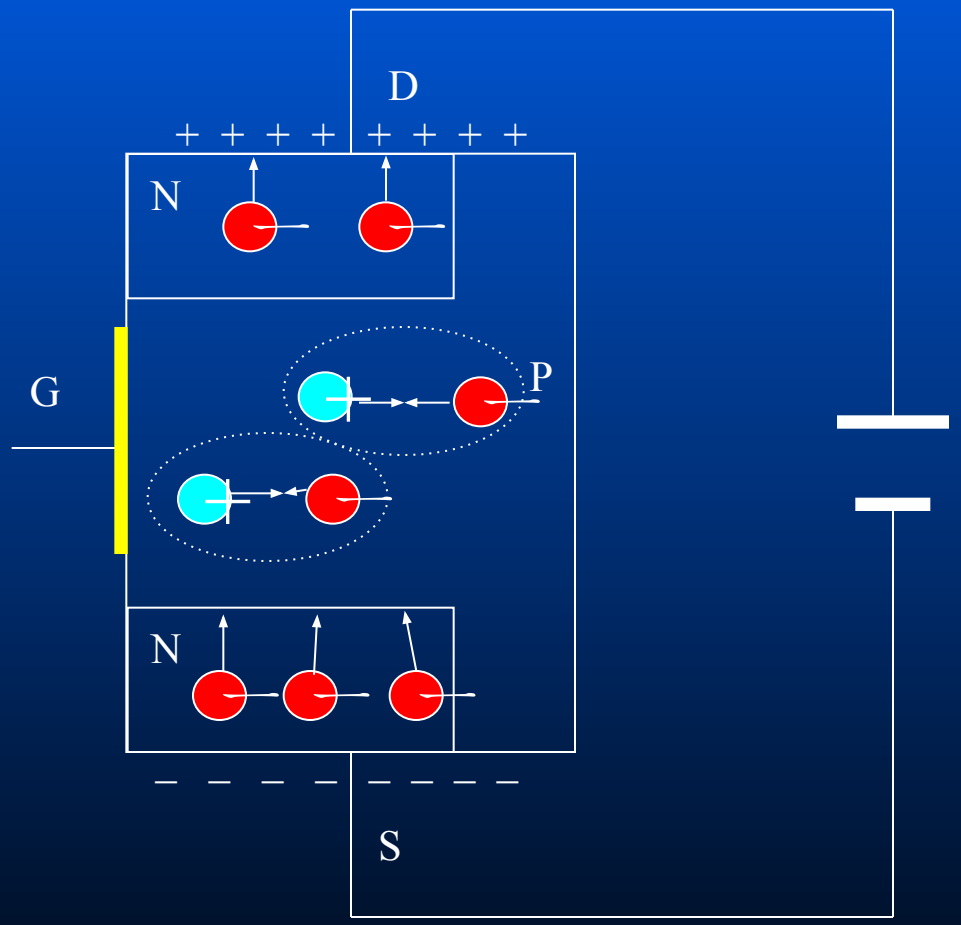


(a) circuit



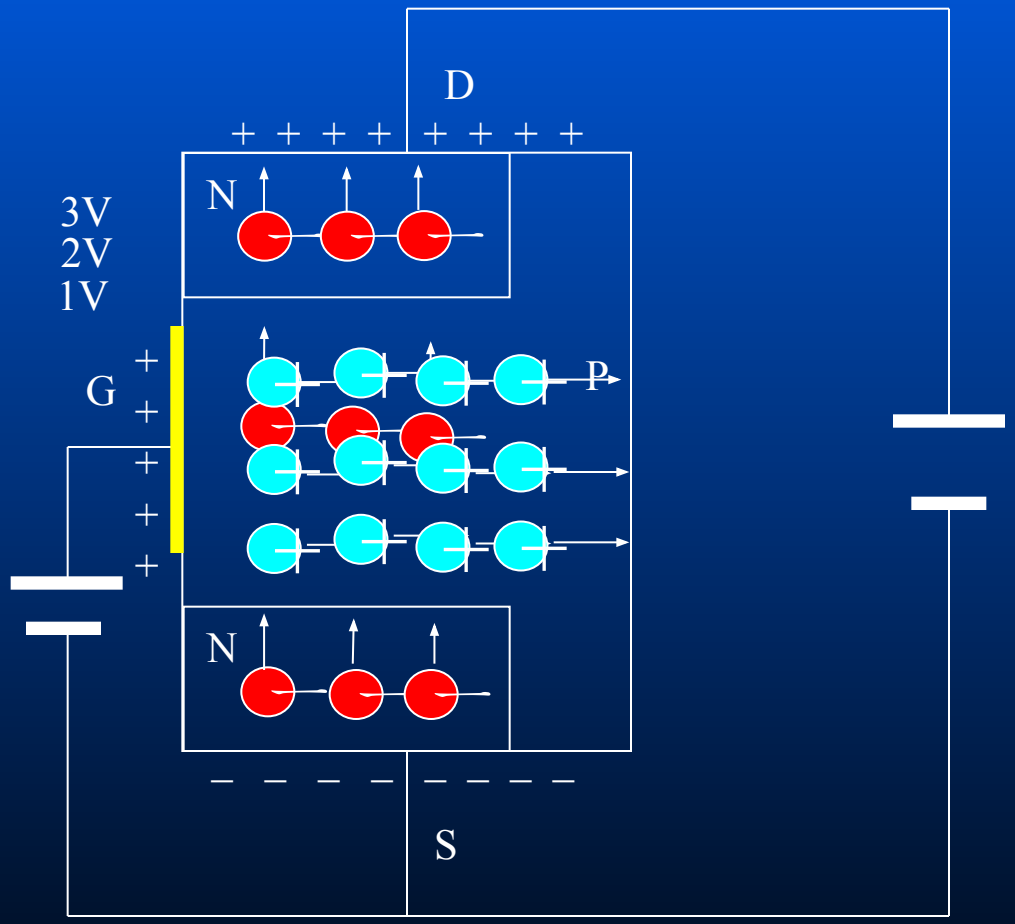
(b) V-I curve

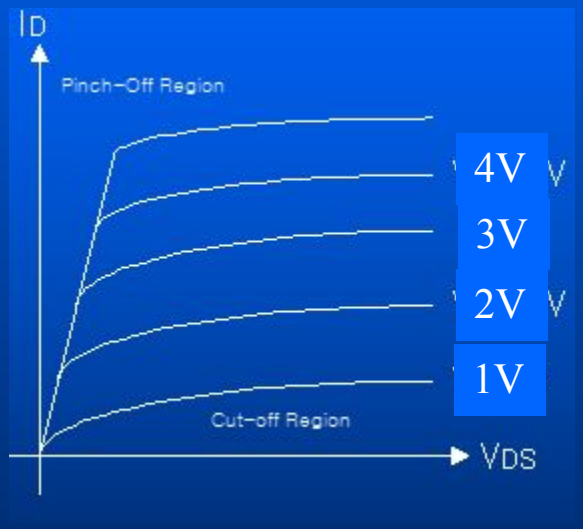
3-6. 증가형 MOSFET의 동작 원리 : Bias 미인가



- GATE 에 전류가 인가되지 않으면
- Source 에서 -가 이동하여도
- Gate의 +와 중화되어
- 전체적으로 전류가 흐르지 못함

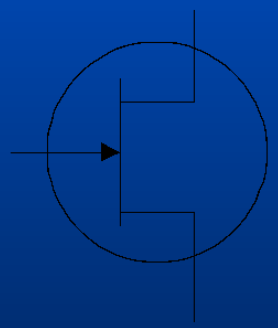
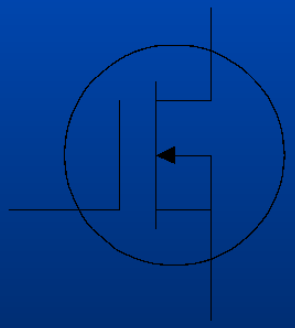
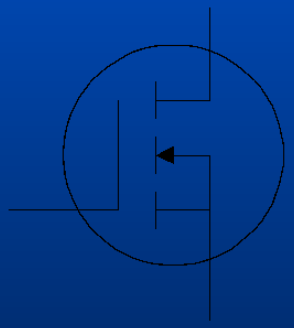
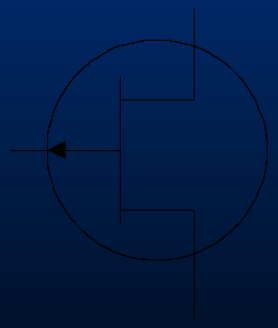
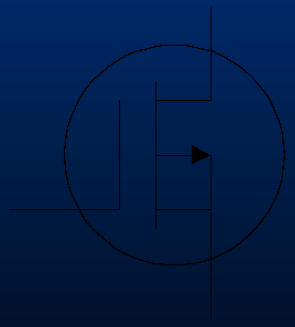
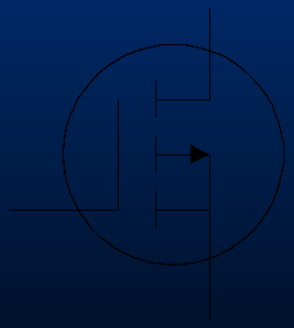
3-5. 증가형 MOSFET의 동작 원리 : Bias 미인가



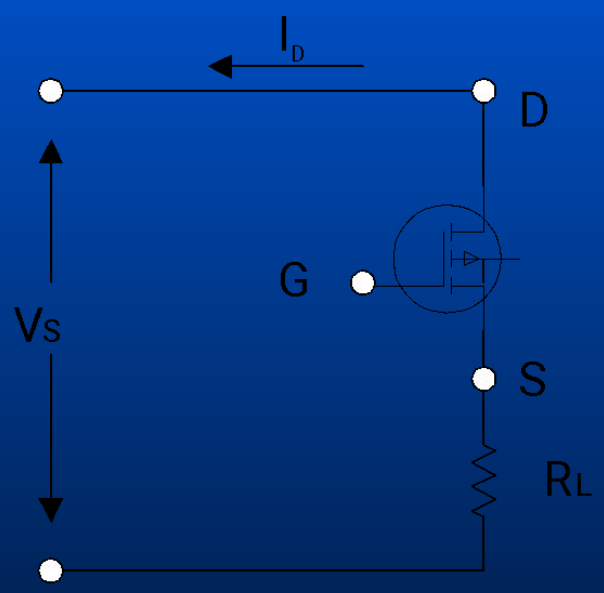


- GATE 에 전압이 인가되면
- Gate에 +을 인가하면 P층의 +와 반발하여 통로가 생김
- Source의 -가 P층의 빈통로로 이동하여 전류가 흐름

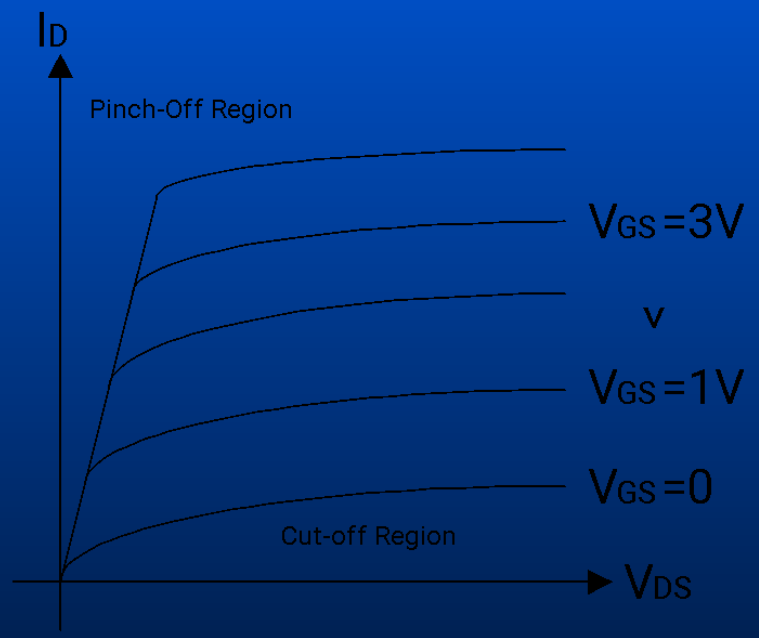
3-9. FET의 동작원리: Symbol

CHANNEL	JFET	□□□ MOSFET	□□□ MOSFET
N			
P			

3-10. FET의 동작원리: 특성 곡선

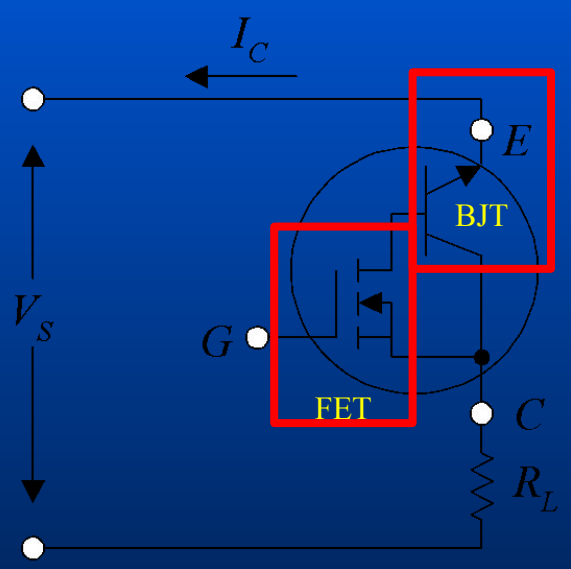


(a) circuit

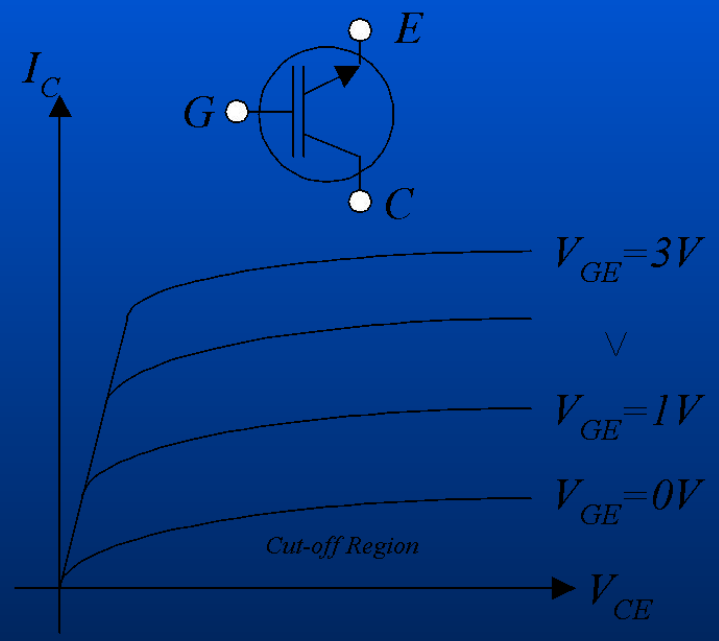


(b) V-I curve

4-1. IGBT(Insulated Gate Bipolar Transistor)



(a) circuit



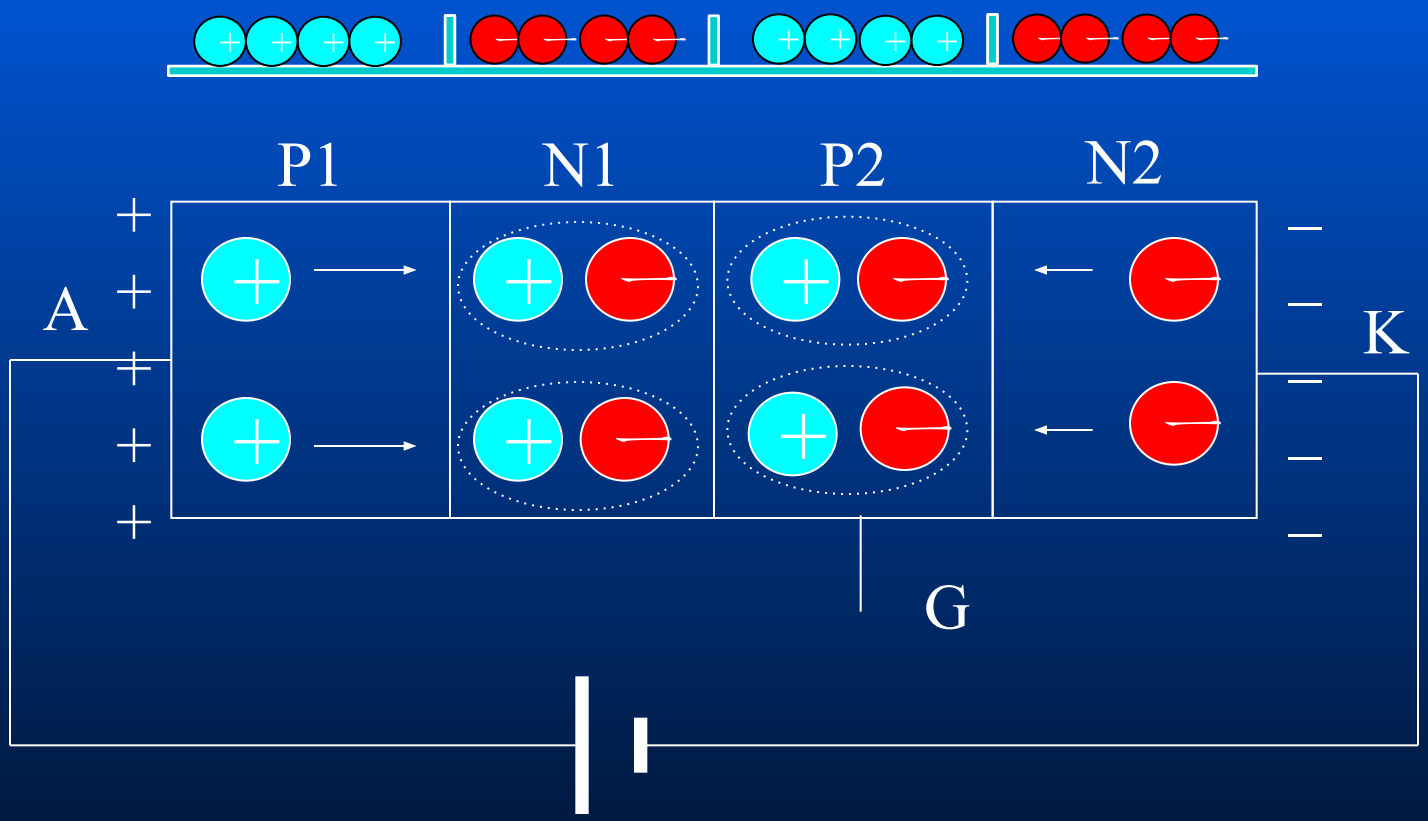
(b) V-I curve

- FET와 BJT의 장점을 이용
- 전압으로 컨트롤, 큰전류 인가 가능

4-2. Transistor 특성

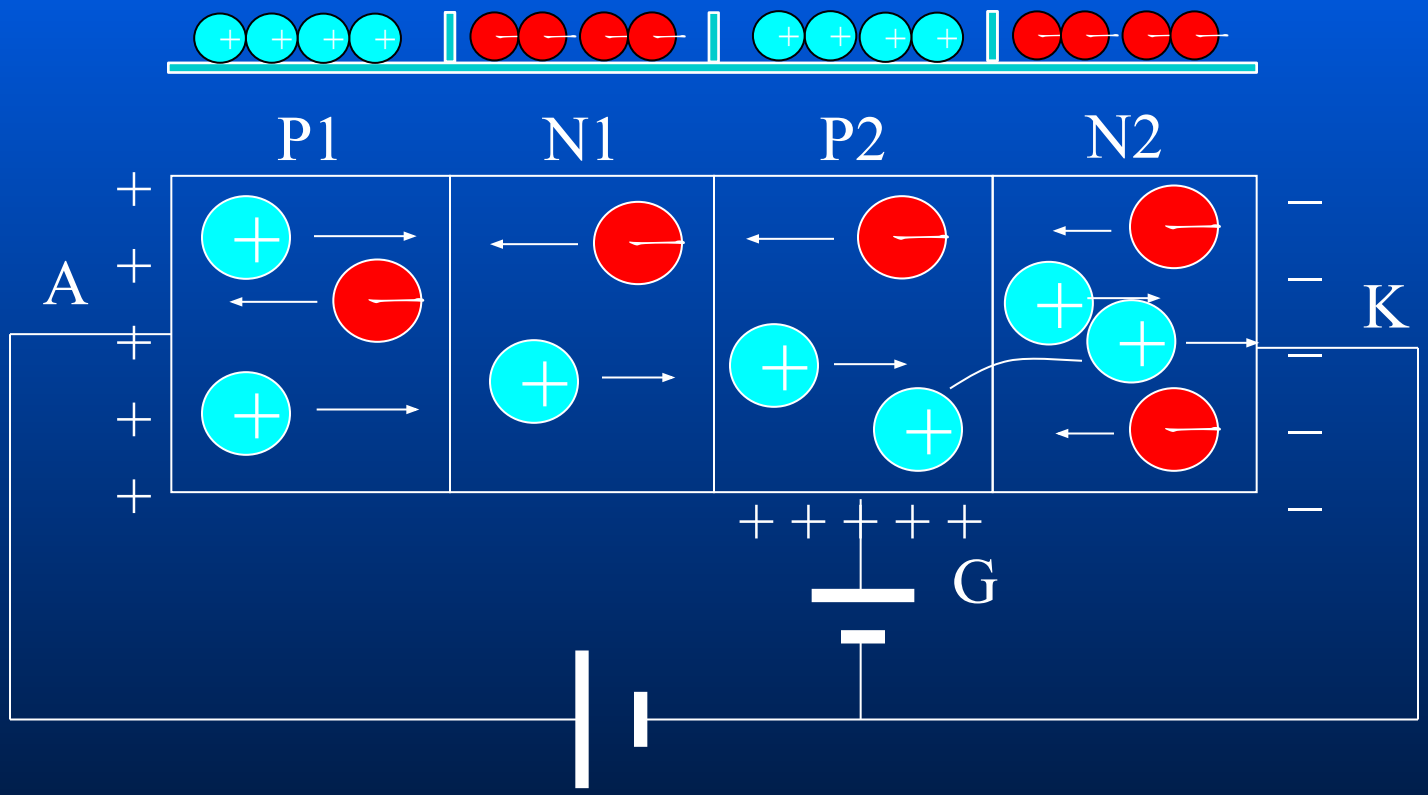
	BJT	FET	IGBT (BJT+FET)
구동방식	전류	전압	전압
인가전류	크다	작다	크다
단가	저가	고가	중간

5-1. SCR(Silicon Controlled Rectifier)



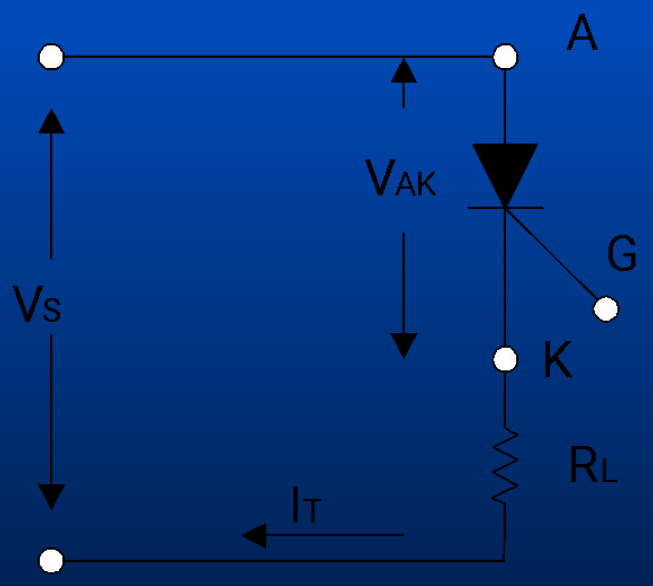
- Anode, Cathode에만 전압을 인가
- N1, P2에서 전자와 정공이 결합 중성화
- 전류가 흐르지 못함

5-2. SCR의 동작원리: Bias 인가

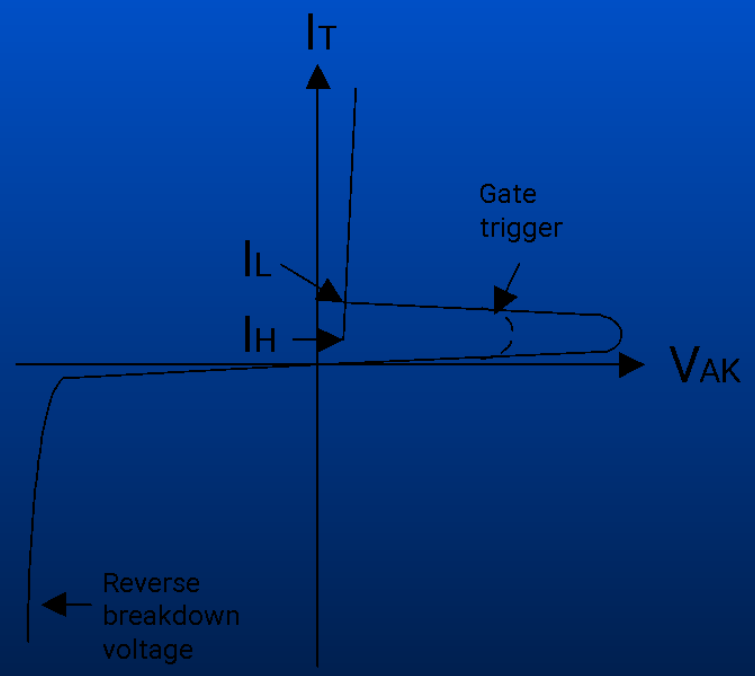


- Gate에 전압이 인가되면 Cathode로 +가 이동
- N2의 -가 P2에 흘러서 통과됨
- N1에는 -만 있으므로 -가 계속 들어오면 넘쳐서 P1으로 이동
- 전체적으로 전류가 흐름

5-4. SCR의 동작원리: 특성곡선



(a) circuit



(b) V-I curve